

Comparitive Performance Report for tentacle-ec-6+2-16k-balance-off vs direct-reads-ec-6+2-16k-balance-off-pr2 vs direct-reads-ec-6+2-16k-balance-reads- on-alexpoc vs tentacle-3-way-replica-balance-reads-on

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Comparison summary for
tentacle-ec-6+2-16k-balance-off vs
direct-reads-ec-6+2-16k-balance-off-pr2 vs
direct-reads-ec-6+2-16k-balance-reads-
on-alex-poc vs
tentacle-3-way-replica-balance-reads-on

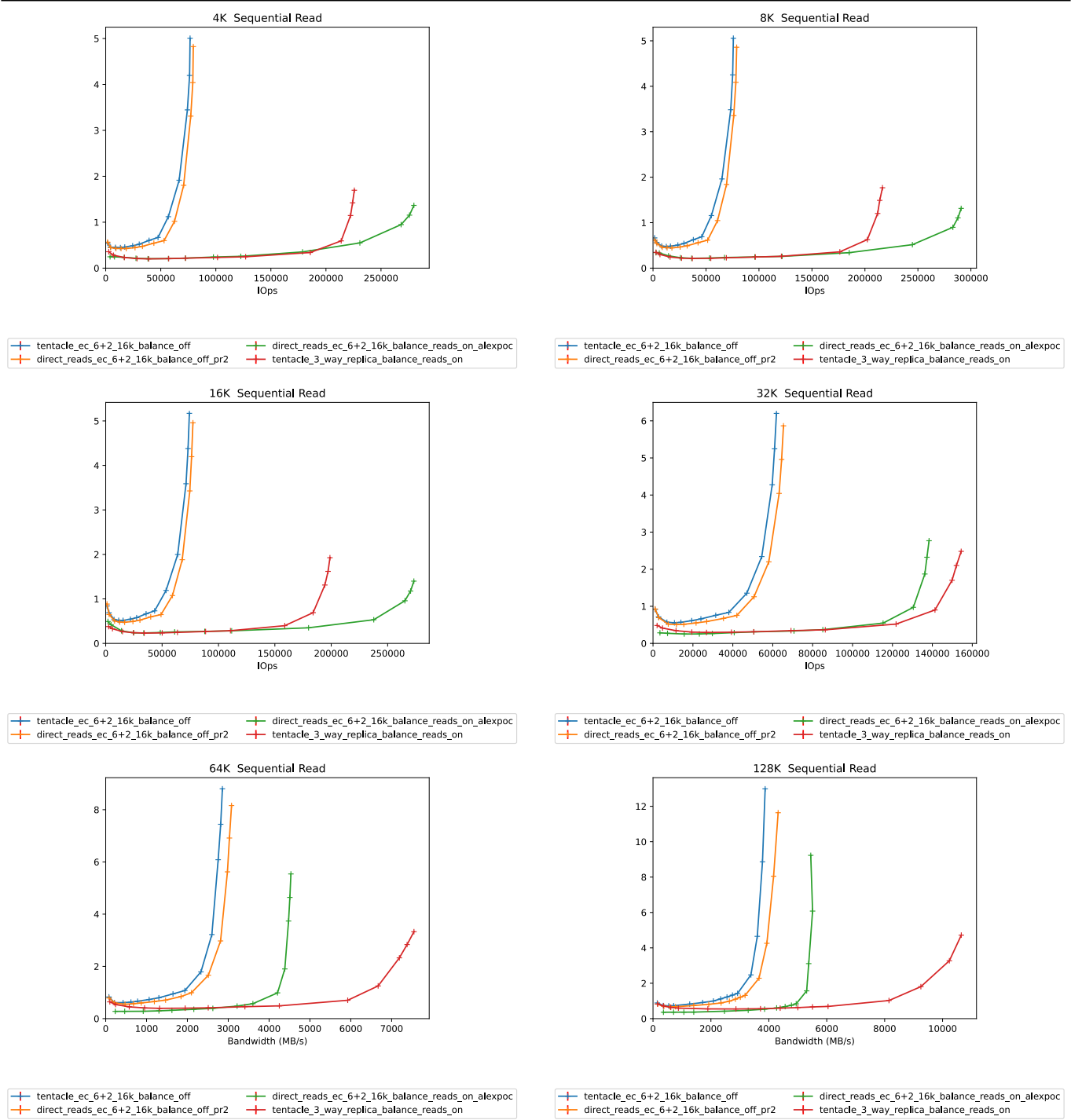
Sequential							
Read	tentacle_ec_6+2-16k-balance-off	direct-reads-ec-6+2-16k-balance-off-pr2	direct-reads-ec-6+2-16k-balance-reads-on-alex-poc	tentacle-3-way-replica-balance-reads-on			
4K	76616@5.0ms	79526@4.8	4%	279868@1.4	265%	225772@1.7	195%
8K	75834@5.1ms	78950@4.9	4%	290830@1.3	284%	216564@1.8	186%
16K	74198@5.2ms	77348@5.0	4%	273262@1.4	268%	198850@1.9	168%
32K	61863@6.2ms	65380@5.9	6%	138344@2.8	124%	154447@2.5	150%
64K	2857@8.8ms	3081@8.2	8%	4535@5.5	59%	7547@3.3	164%
128K	3873@13.0ms	4321@11.6	12%	5511@6.1	42%	10647@4.7	175%
256K	5386@18.7ms	6155@16.3	14%	8710@9.6	62%	12908@7.8	140%
384K	6063@33.2ms	7055@28.5	16%	10177@9.9	68%	12866@7.8	112%
512K	6760@39.7ms	7813@34.4	16%	11287@11.9	67%	12804@5.2	89%
768K	7129@56.5ms	8316@48.4	17%	12053@16.7	69%	12380@8.1	74%
1024K	7578@53.2ms	8791@45.8	16%	12895@5.2	70%	12189@5.5	61%
2048K	7633@52.8ms	8866@45.4	16%	13207@5.1	73%	10251@13.1	34%
4096K	7388@72.7ms	8605@62.4	16%	12653@10.6	71%	10070@13.3	36%

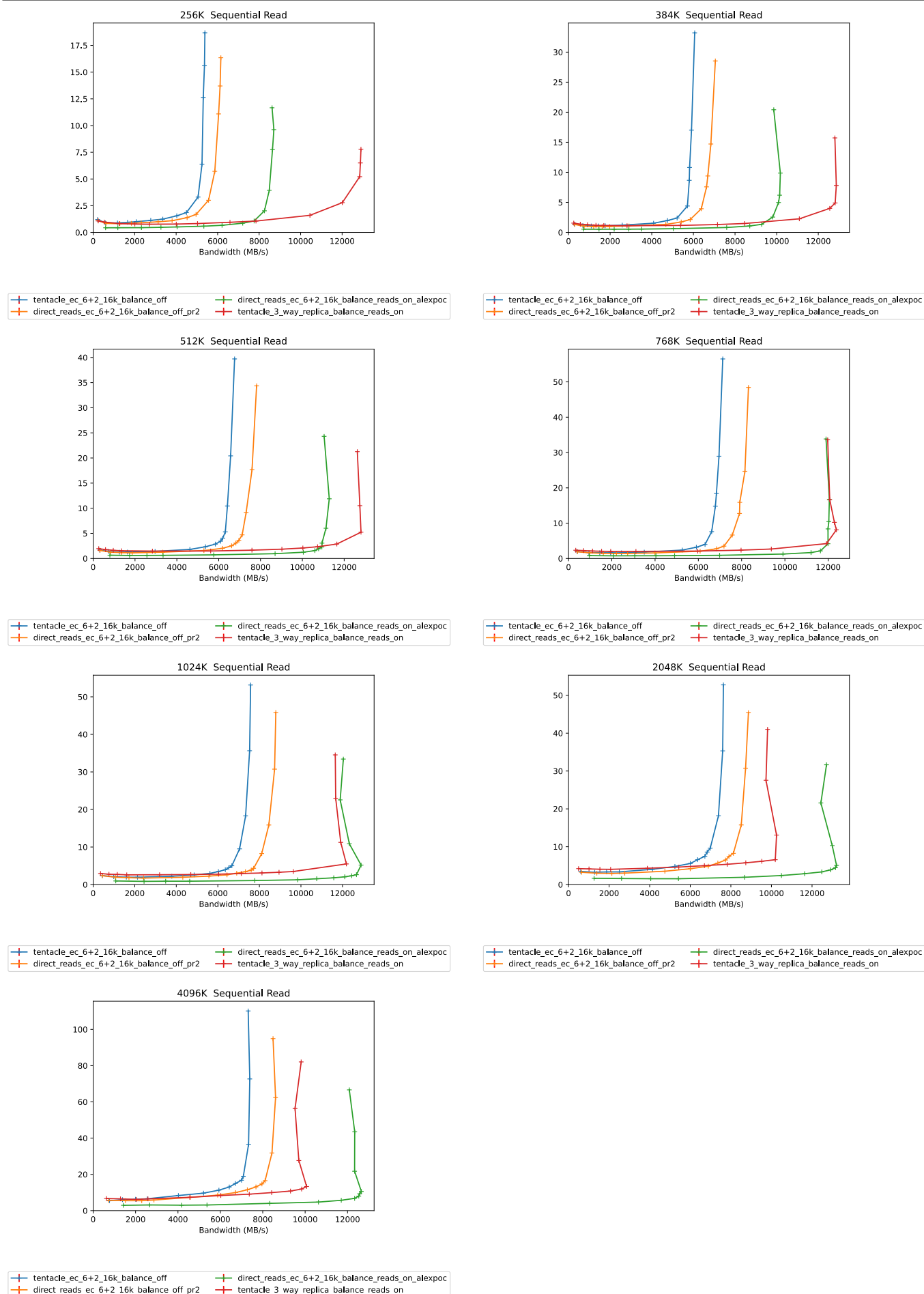
Random							
Read	tentacle_ec_6+2-16k-balance-off	direct-reads-ec-6+2-16k-balance-off-pr2	direct-reads-ec-6+2-16k-balance-reads-on-alex-poc	tentacle-3-way-replica-balance-reads-on			
4K	67430@5.7ms	81507@4.7	21%	204454@1.9	203%	171696@2.2	155%
8K	65784@5.8ms	79399@4.8	21%	196721@1.9	199%	161660@2.4	146%
16K	63515@6.0ms	76426@5.0	20%	184396@2.1	190%	152226@2.1	140%
32K	47329@8.1ms	55337@6.9	17%	91855@4.2	94%	133590@2.9	182%
64K	2103@10.0ms	2376@8.8	13%	3013@8.3	43%	7059@3.0	236%
128K	3060@16.4ms	3410@14.8	11%	3862@13.0	26%	10382@3.2	239%
256K	4713@21.3ms	5498@18.3	17%	6328@15.9	34%	12336@2.7	162%
384K	5389@37.4ms	6214@32.4	15%	7799@25.8	45%	12192@4.1	126%
512K	5912@45.4ms	6986@38.4	18%	9343@28.7	58%	12491@5.4	111%
768K	6448@62.5ms	7376@54.6	14%	10383@38.8	61%	12014@4.2	86%
1024K	6987@57.6ms	8010@50.3	15%	11586@5.8	66%	12103@5.5	73%
2048K	7391@36.3ms	8426@47.8	14%	12773@5.2	73%	10678@12.5	44%
4096K	7410@108.7ms	8475@63.3	14%	12386@8.1	67%	10171@13.2	37%

Random							
Read/Write	tentacle_ec_6+2-16k-balance-off	direct-reads-ec-6+2-16k-balance-off-pr2	direct-reads-ec-6+2-16k-balance-reads-on-alexpoc	tentacle-3-way-replica-balance-reads-on	tentacle_ec_6+2-16k-balance-off	direct-reads-ec-6+2-16k-balance-off-pr2	direct-reads-ec-6+2-16k-balance-reads-on-alexpoc
4K_70/30	33168@11.6ms	40274@9.5	21%	56005@6.8	69%	86455@4.4	161%
16K_70/30	31252@12.3ms	37486@10.2	20%	51593@7.4	65%	78340@3.3	151%
64K_70/30	1208@20.8ms	1372@18.3	14%	1521@16.5	26%	3886@5.4	222%
64K_30/70	887@28.3ms	1004@25.0	13%	1058@23.7	19%	2662@9.4	200%

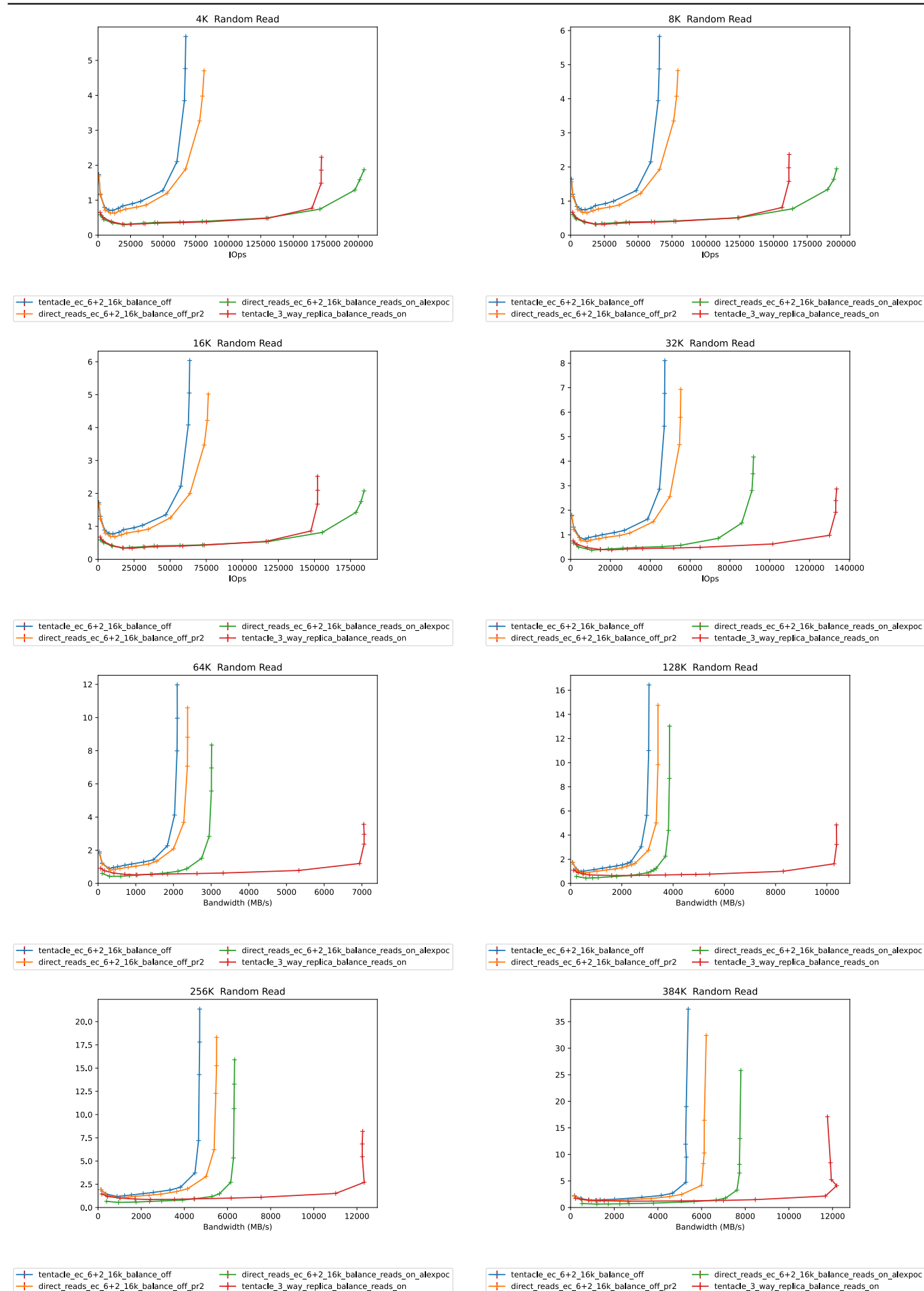
Response Curves

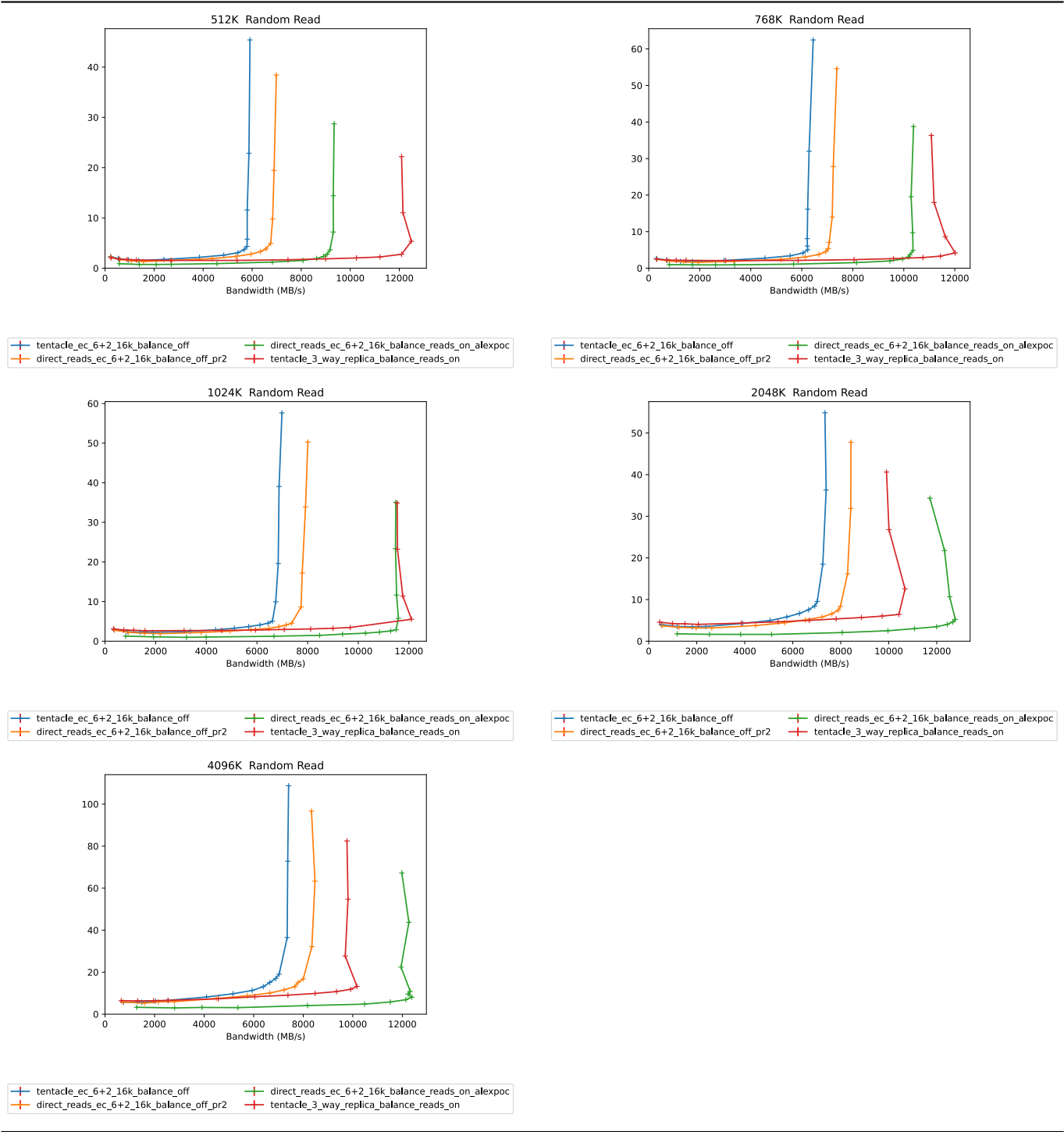
Sequential Read



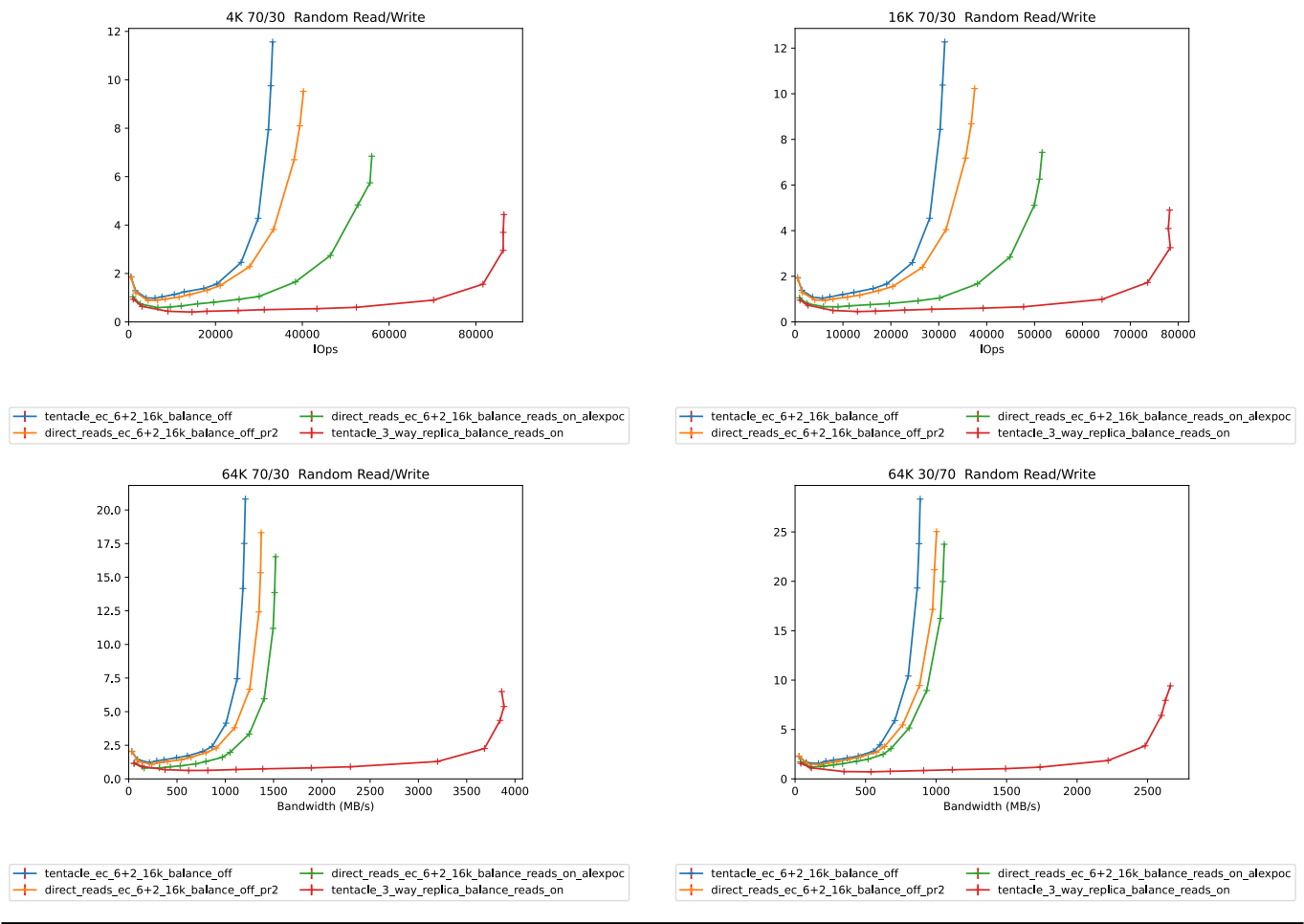


Random Read





Random Read/Write



Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```

- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4

```

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

32krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 32768
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

32ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 32768
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

384krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 393216
- total_iodepth:
- 1

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512

384ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 393216
- total_iodepth:
- 1
- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512

4Mrandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4194304
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4Msequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4194304
- total_iodepth:
- 1

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 4096
- rwmixread: 70
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4096
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4096
- total_iodepth:

```

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30

```

```

    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64kmixread70:
    jobname: mixread
    mode: randrw
    numjobs:
      - 1
    op_size: 65536
    rwmixread: 70
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1
    op_size: 65536
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1

```

```

  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph_mgr_cmd: /usr/bin/ceph-mgr
  ceph_mon_cmd: /usr/bin/ceph-mon
  ceph_osd_cmd: /usr/bin/ceph-osd
  ceph_run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```

    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
osds_per_node: 6
pdsh_ssh_args: -a -x -l%u %h
rados_cmd: /usr/bin/rados
rbd_cmd: /usr/bin/rbd
tmp_dir: /tmp/cbt
use_existing: true
user: cjames
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -FO --rawtoo --sep ";" -f {collectl_dir}

```